Course Code	**EMT22020**					
Course Category	**Program Foundation**					
Course Title	**Calculus and Numerical Methods (CNM)**					
Teaching Scheme	**Lectures**	**Tutorials**	**Laboratory/ Practical**	**Project**	**Total**
Weekly load hours	**3**	**1**	**-**	**-**	**4**
Credits	**3**	**1**	**-**	**-**	**4**
Assessment Schema Code	**TT1**				

Pre-requisites: HSC Mathematics, Linear Algebra and Differential Calculus

Course Objectives:

- To learn linear differential equations and their applications in Engineering.
- To learn partial differential equation and their applications.
- To understand Laplace transform techniques and solve differential equations using Laplace transforms.
- To understand Fourier transforms techniques and their applications.
- To learn numerical methods for solving algebraic & differential equations.

Course Outcomes:

After completion of this course students will be able to

1. identify first order & higher order linear differential equations & solve these equations using various methods. (CL I & III)
2. understand the concept of partial differential equations used in boundary value problems & solve heat equation & wave equation using the method of separation of variables (CL II & III)
3. apply Laplace transform techniques to solve ordinary differential equations. (CL III)
4. apply Fourier transform techniques to solve differential equations involved in real life engineering problems. (CL III).
5. understand numerical methods for differentiation and integration processes (CL II & III)
6. solve ordinary differential equations using various numerical methods (CL III)

Course Contents:

Unit 1: Linear Differential Equation:
Review of first order differential equations, Linear Differential Equation of n^{th} order with constant coefficients, Method of variation of parameters, Cauchy's and Legendre's Differential Equations, Applications of Linear differential equations-mass spring systems and bending of beams problems.

Unit 2: Partial Differential Equations:
Basic concepts, Solution of Partial Differential equations, method of separation of variables for the solution of one- and two-dimensional Heat flow equations, Wave equation.

Unit 3: Laplace Transform:
Definition, Properties, Laplace Transform of standard functions, Laplace transform of some special functions, Inverse Laplace Transform, Applications of Laplace Transform for solving Ordinary differential equations.

Unit 4: Fourier Transform:
Introduction of Fourier series, half range sine & cosine series, Fourier Integral theorem, Fourier Sine and Cosine Transforms, Inverse Fourier Transform. Finite Fourier Transform, Applications of Fourier transforms to problems on one- and two-dimensional heat flow problems.

Unit 5: Numerical Interpolation, Differentiation and Integration:
Calculus of Finite difference, Finite difference Operators, Newton's Forward and Backward Interpolation, Lagrange's Interpolation formula. Numerical differentiation & Numerical Integration, Trapezoidal Rule, Simpson's $1/3^{rd}$and $3/8^{th}$ rules, Error analysis.

Unit 6: Numerical solution of ordinary differential equations:
Ordinary differential equations, initial value problems, Euler's method, modified Euler's method, Runge–Kutta method Predictor-Corrector method.

Tutorial Exercises:

1. Linear Differential Equations solution by Shortcut method
2. General, Variation of Parameter methods
3. Applications of Linear Differential Equations.
4. Wave equation,
5. One dimensional Heat flow equations.
6. Two-dimensional Heat flow equations
7. Laplace transform of standard functions
8. Inverse Laplace Transform
9. Solve ODE using Laplace transform

10. General Fourier transform
11. Fourier Sine and Cosine Transforms.
12. Problems on Numerical interpolation and differentiation
13. Problems on Numerical integration.
14. Problems on Taylor's series & Euler's method
15. Problems on Runge kutta & predictor-corrector methods

Note: Introduce Mathematical Software for few tutorial conduction. Tutorial shall be engaged in 3 batches (batch size of 20 students) per division.

LearningResources:

Reference Books:

1. KreyszigErwin, "Advanced Engineering Mathematics" 10th edition ,Wiley Eastern Limited 2015.
2. Greenberg Michael D.,"Advanced Engineering Mathematics", 2nd edition, Pearson . 2009.
3. Grewal B.S., "Higher Engineering Mathematics" ,43rd edition Khanna Publishers 2014
4. ChapraS.C.and CanaleR.D.," Numerical Methods for Engineers",WCB/McGraw-Hill Publications, 2001

Supplementary Reading:

1. O' Neil Peter, "Advanced Engineering Mathematics", 8th edition, Cengage Learning 2015.
2. Weber H.J. and Arfken G.B. "Mathematical Methods For Physicists" , 6th edition, Academic Press 2011.
3. Maurice D Weir,Joel Hass, Frank R Giordano, " Thomas' Calculus" , 14th edition, Pearson 2009.

Web Resources:

Web links:

- Introduction to second order LDE https://www.youtube.com/watch?v=tGtCajxHoDw
- Fourier Transform, Fourier Series, and frequency spectrumhttps://www.youtube.com/watch?v=r18Gi8lSkfM

MOOCs: NPTEL, MIT OPEN COURSEWARE

- https://ocw.mit.edu/courses/mathematics/18-02sc-multivariable-calculus-fall-2010/
- https://ocw.mit.edu/courses/mathematics/18-03-differential-equations-spring-2010/video-lectures/lecture-9-solving-second-order-linear-odes-with-constant-coefficients/
- http://nptel.ac.in/courses/111103021/18

- *https://ocw.mit.edu/courses/mathematics/18-02-multivariable-calculus-fall-2007/video-lectures/lecture-8-partial-derivatives/*

Pedagogy:

- Team Teaching
- Tutorials and class tests/assignments
- Audio- Video technique